

IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM Cloud”

Project Title – Rural Infrastructure Project Classification Using Machine Learning

Domain of Project – Rural Development / Infrastructure

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Problem Statement

Problem Statement: Rural Infrastructure Project Classification Using Machine Learning

Rural infrastructure development plays a vital role in improving transportation, education, healthcare, and economic growth. Government projects are implemented under different PMGSY (Pradhan Mantri Gram Sadak Yojana) schemes based on project characteristics. Manually classifying these infrastructure projects is time-consuming and prone to errors. This project aims to develop a Machine Learning model that automatically classifies rural infrastructure projects into the appropriate PMGSY category using historical project data. The solution helps improve decision-making, reduces manual effort, and supports efficient project management.

Dataset link :

https://aikosh.indiaai.gov.in/web/datasets/details/pradhan_mantri_gram_sadak_yojna_pmgsy.html

Proposed Solution - IBM watsonx.ai Studio – AutoAI

This project uses IBM watsonx.ai Studio AutoAI to automate the classification of rural infrastructure projects.

The solution performs:

- Data preprocessing
- Automatic feature engineering
- Machine Learning model selection
- Hyperparameter optimization
- Model evaluation

The trained model predicts the correct PMGSY project category with high accuracy, enabling faster and more reliable classification.

Technology Used

Platform

- IBM Cloud
- IBM watsonx.ai Studio
- IBM AutoAI

Programming Language

- Python

Machine Learning Type

- Classification

Algorithm Used

- Snap Random Forest Classifier
- XGBoost Classifier

Technology Used

Dataset - PMGSY_DATASET.csv

https://aikosh.indiaai.gov.in/web/datasets/details/pradhan_mantri_gram_sad_ak_yojna_pmgsy.html

Accuracy Obtained :

The Snap Random Forest Classifier was selected as the best model with an accuracy of **92.4%**.

Hyperparameter Tuning :

Automatically optimized using IBM AutoAI

Training Screenshots-

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Experiment summary

Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score

Progress map ⓘ
Prediction column: PMGSY_SCHEME

The progress map illustrates the experimental workflow. It begins with a linear sequence of steps: 'Read dataset', 'Split holdout data', 'Read training data', 'Preprocessing', and 'Model selection'. From 'Model selection', the workflow branches into two parallel paths. The top path consists of 'Snap Random Forest Classifier', 'Hyperparameter optimization' (P1), 'Feature engineering' (P2), 'Hyperparameter optimization' (P3), and 'Ensemble creation' (P4). The bottom path consists of 'XGB Classifier', 'Hyperparameter optimization' (P6), 'Feature engineering' (P7), 'Hyperparameter optimization' (P8), and 'Ensemble creation' (P9). Both paths converge at a final 'Model selection' step (P10), which is highlighted with a star icon.

Relationship map

Swap view ↺

The relationship map provides a visual overview of the connections between the steps in the progress map. It shows a semi-circular arrangement of nodes representing the steps, with lines indicating the flow and dependencies between them. The parallel paths from 'Model selection' are clearly visible, along with their convergence at the final 'Model selection' step.

Experiment completed ✓
10 PIPELINES GENERATED
10 pipelines generated from algorithms. See pipeline leaderboard below for more detail.
Time elapsed: 7 minutes

View log

Save code

Training Screenshots-

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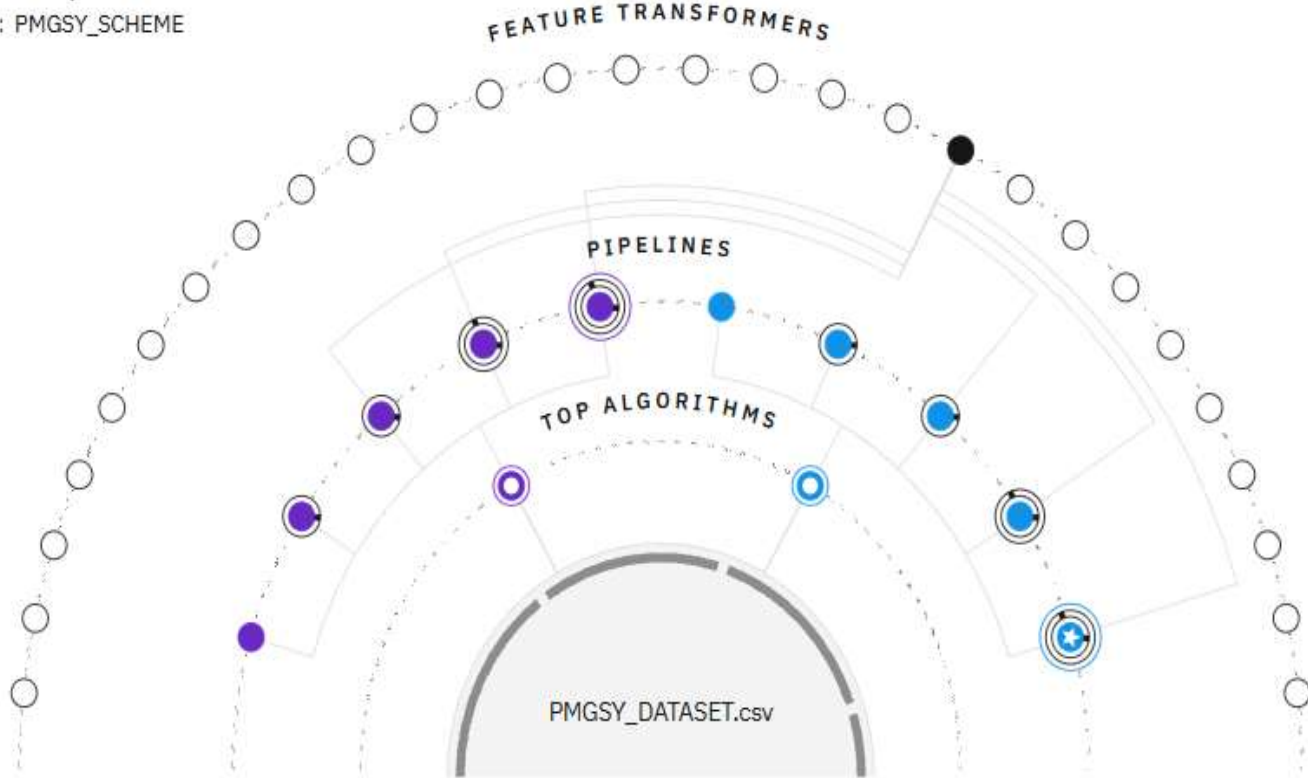
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Experiment summary

Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score ⚙

Relationship map ⓘ
Prediction column: PMGSY_SCHEME



Progress map
[Swap view](#) ⇄

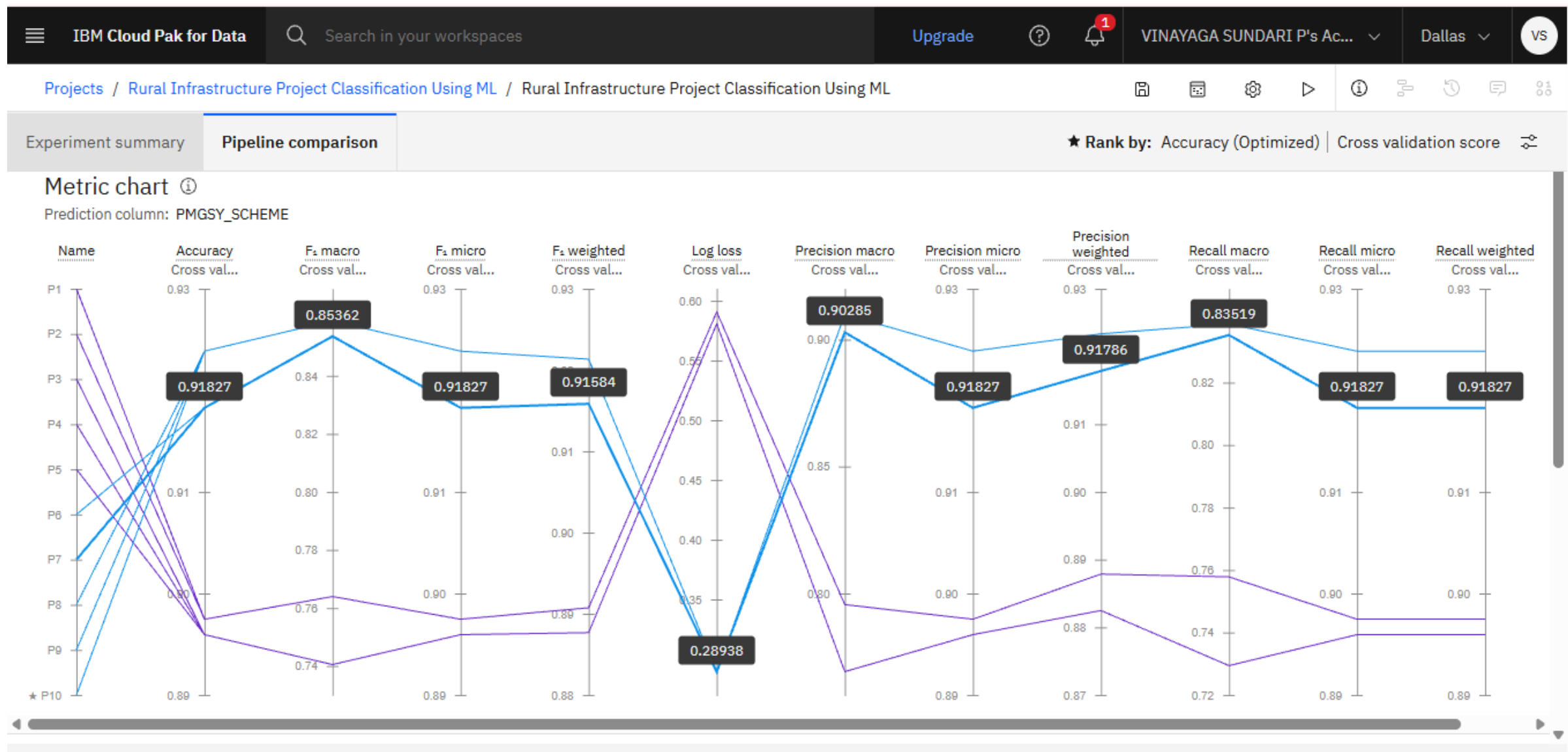


Experiment completed ✓
10 PIPELINES GENERATED
10 pipelines generated from algorithms. See pipeline leaderboard below for more detail.
Time elapsed: 7 minutes

[View log](#)

[Save code](#)

Training Screenshots-



Best Model Screenshot:

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Experiment summary

Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score

time elapsed: 7 minutes

View log

Save code

Pipeline leaderboard

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	Rank ↑	Name	Algorithm	Specialization	Accuracy (Optimized) Cross validation	Enhancements	Build time
★	1	Pipeline 10	Batched Tree Ensemble Classifier (XGB Classifier)	INCR	0.924	HPO-1 FE HPO-2 BATCH	00:02:18
	2	Pipeline 9	XGB Classifier		0.924	HPO-1 FE HPO-2	00:02:10
	3	Pipeline 8	XGB Classifier		0.924	HPO-1 FE	00:01:21
	4	Pipeline 7	XGB Classifier		0.918	HPO-1	00:00:23

Testing Screenshot:

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Rural Infrastructure Project Classification Using ML

Deployed

Online

API reference

Test

Evaluations

Transactions

Enter input data

Text

JSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

Clear all

	STATE_NAME (other)	DISTRICT_NAME (other)	NO_OF_ROAD_WORK_SANCTIONED (integer)	LENGTH_OF_ROAD_WORK_SANCTIONED (double)	NO_OF_BRIDGES
1	Andhra Pradesh	Anantapur	619	2169.05	35
2	Andhra Pradesh	Chittoor	18	26.045	0
3	Andhra Pradesh	Guntur	13	141.44	1

3 rows, 14 columns

Predict

Testing Screenshot:

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Prediction results

Close

X

Prediction type

Multiclass classification

Prediction percentage



3 records

Display format for prediction results

☒ Table view ☐ JSON view

☐ Show input data ⓘ

	Prediction	Confidence
1	PMGSY-I	100%
2	PMGSY-II	86%
3	PMGSY-II	87%
4		
5		
6		
7		
8		

Download JSON file

Novelty and Uniqueness

- Automates rural infrastructure project classification.
- Reduces manual classification errors.
- Uses IBM AutoAI for automatic model generation.
- Provides fast and accurate predictions.
- Supports government planning and rural development initiatives.
- Easily scalable for larger infrastructure datasets.

Git Hub Link

Please create public github repository (Enable Add README button while creating repository) with format –

Please test and Paste the working Github URL –

Make sure you have Uploaded following three files into your github repository

1) Data.csv file, 2) yourproblemstatement.pdf file 3) your project presentation.pptx file 4) Python Notebook downloaded from IBM Cloud

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Future Scope

- Improve prediction accuracy using larger datasets.
 - Integrate GIS and satellite data for infrastructure planning.
 - Develop a web application for real-time project classification.
 - Deploy the model as a cloud-based API using IBM Cloud.
- Extend the solution to classify additional government infrastructure schemes.

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Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.